

Deep research report on CATE, latent tags, and validation signals in your causal-irregular-time-series results

Scope and evidence base

This report analyzes the *provided* final outputs (from `final_results.zip`) for the two datasets in your pipeline: **PhysioNet 2012-style ICU time series** (7,993 aligned patients in the final majority-vote cohort; baseline in-hospital mortality $\approx 14.24\%$) and **MIMIC-III-style ICU stays** (26,845 aligned stays; baseline in-hospital mortality $\approx 12.08\%$). The analysis follows the workflow in your diagram: clinically motivated latent tags $\rightarrow$ (multi-model) latent tagging + majority vote $\rightarrow$ latent validation by mortality prediction $\rightarrow$ causal inference (CausalForestDML/LinearDML/matching) $\rightarrow$ validation (sensitivity/robustness + permutation tests + falsification concepts).

A key interpretability constraint must be stated up front. Your “treatments” are **latent clinical states** (e.g., Shock, RenalDysfunction), not randomized interventions. In the potential-outcomes framework, causal effects require *well-defined treatment versions* (a core part of SUTVA/consistency) so that counterfactual outcomes $Y(a)$ are meaningful. When an exposure has many versions (or is not realistically manipulable), “the causal effect of the exposure” becomes less well-defined and can mix multiple mechanisms. ¹ This matters especially for syndromic/diagnostic constructs like “Shock” that can arise from distinct etiologies and clinical trajectories (sepsis, hemorrhage, cardiogenic shock, sedation/measurement artifacts, etc.), which can complicate interpretation even if a model produces stable-looking numbers. ²

Accordingly, where I say “effect,” interpret it as: **the model-estimated change in mortality risk under a hypothetical intervention that toggles the latent tag, conditional on the chosen adjustment strategy**—not as a clinical claim that “shock itself doesn’t cause death” or that “renal failure causes exactly X% mortality.” The goal here is to judge whether your outputs show **plausible directional incentives and consistency with external clinical evidence**, and where they raise red flags.

Workflow traceability and what your artifacts empirically show

Your workflow diagram describes a modular pipeline: preprocessed ICU time series $\rightarrow$ latent variables + outcomes + domain DAG $\rightarrow$ multi-voter tagging (rules + neural models including STraTS) $\rightarrow$ majority-vote latent dataset $\rightarrow$ latent validation via outcome prediction $\rightarrow$ causal inference module $\rightarrow$ CATE outputs $\rightarrow$ CATE validation via sensitivity + permutations + falsification.

Within your *zip outputs*, several concrete traceability points are important:

- **Majority vote alignment step is selection-defining:** both datasets are reduced to the **intersection** of patients/stays that exist in *all* voter files (rule-based reference + GRU + GRU-D + TCN + STraTS). This yields 7,993 (PhysioNet) and 26,845 (MIMIC) final rows. This helps consistency of downstream analyses, but also creates a **selection filter** (the final cohort is “patients for whom all tagging models produced predictions”), which can in principle shift prevalences and

outcomes versus the full cohort. This is a real-world source of external validity shift (not “bias” per se, but a change in target population).

- **Mortality prediction from latent tags is moderate, not near-perfect**, indicating that the latent tags carry signal but do not fully summarize risk. That is *good* for avoiding trivial leakage, but it also means residual confounding/omitted severity factors remain plausible.
- **Cardiac injury treatment is effectively unusable in your final vote**: the rule-based “CardiacInjury/CardInj” label is **all zeros** in both datasets, and majority vote therefore yields all zeros as well (regardless of STraTS behavior). This prevents CATE estimation for that treatment and is a major diagnostic about label construction (details below).

These observations set the stage for interpreting your CATE and validation diagnostics, because they define: (i) the cohort, (ii) label quality, and (iii) the plausible amount of remaining unmeasured confounding.

Latent variable construction and tagging performance

Clinical grounding of latent tag definitions

Even without treating the tags as fully “intervenable,” their *clinical meaning* matters. Many of your tags align closely with **established ICU organ dysfunction constructs**, especially **SOFA** components and sepsis/shock physiologic thresholds:

- **Renal dysfunction / failure**: SOFA’s renal component uses creatinine cut points including **≥2.0 mg/dL** and urine output cut points including **<500 mL/day**, which match the style of thresholds used in your renal tagging logic and directly connect renal dysfunction states to validated severity scoring. ³
KDIGO’s AKI definition relies on creatinine rise and urine output criteria (e.g., oliguria thresholds), providing additional clinical legitimacy for a kidney-injury construct derived from creatinine/urine output patterns. ⁴
- **Hepatic dysfunction**: SOFA’s liver component uses bilirubin brackets with **≥2.0 mg/dL** as a clinically meaningful threshold for dysfunction severity. ³
- **Respiratory failure**: the **Berlin ARDS definition** classifies ARDS severity using **PaO₂/FiO₂** cutoffs including **≤300**, supporting your use of oxygenation indices as respiratory dysfunction proxies. ⁵
- **Shock**: modern sepsis/shock frameworks emphasize **MAP targets around 65 mmHg** and lactate as a perfusion marker. The Surviving Sepsis guidelines recommend MAP targets around **65 mmHg** in septic shock resuscitation, and Sepsis-3 uses *vasopressor requirement to maintain MAP ≥65 plus lactate >2 mmol/L* as a clinical identifier for septic shock. ⁶
Your shock tag is broader than Sepsis-3 (because you do not explicitly encode vasopressor use “despite fluids”), but the physiologic anchors are consistent with the ICU literature.
- **Inflammation / infection-like physiology**: classic SIRS criteria include temperature and WBC thresholds (e.g., **T >38°C or <36°C; WBC >12k or <4k**, etc.). While Sepsis-3 moved away from SIRS as the defining criterion, it explicitly notes SIRS-type findings remain useful as *nonspecific supportive evidence* of infection/inflammation. ⁷

This matters because it means your latent variables are not arbitrary embeddings: they are **clinically interpretable organ dysfunction indicators** that align with established scoring and definitions. That,

in turn, makes it *plausible* (not guaranteed) that their relationship to mortality (association and potentially causal contribution) is directionally positive for most organs.

Model-based tag prediction and “why latents are not random”

Your STraTS output summaries show strong overall discrimination for the latent tag prediction task (aggregated metrics), and you include multiple time-series architectures designed for missingness and irregular sampling:

- **GRU-D** was explicitly proposed to handle multivariate time series with missing values, incorporating masking and time interval information (“informative missingness”). ⁸
- **STraTS** is specifically designed for sparse, irregular clinical time series by representing observations as triplets (time, variable, value) and using self-supervision to learn representations without discretization. ⁹
- **DML/forest-based CATE estimation’s first-stage modeling** is compatible with using rich ML models for nuisance functions, but the tagger stage itself is separate: it is your *labeling* layer.

From your training summaries:

- **MIMIC latent tagging** (test): STraTS performs best by AUROC (≈ 0.938) and AUPRC (≈ 0.962) compared with GRU-D, SAND, GRU, and TCN.
- **PhysioNet latent tagging** (test): GRU-D performs best by AUROC (≈ 0.937) and AUPRC (≈ 0.950), with STraTS slightly lower (AUROC ≈ 0.925 , AUPRC ≈ 0.939).

These results support a narrow but important claim: **your latent tags are learnable from the time series**, and the learned models can reproduce them with high discrimination—consistent with the idea that the tags reflect coherent physiological patterns rather than pure noise. ¹⁰

Majority vote: justification and a critical caveat

Your majority vote combines **five voters** (rule-based reference + 4 neural models). Majority voting is a standard ensemble approach and can improve robustness by reducing variance and canceling individual model errors when errors are not perfectly correlated. ¹¹

It is also conceptually related to weak supervision/data programming ideas: multiple imperfect labeling sources can be combined to yield better training labels than any single heuristic source. ¹²

Critical caveat from your own outputs: the “CardiacInjury/CardInj” label in your rule-based output is **0% prevalence** (all zeros) for both datasets, which implies the majority vote **must** be all zeros too (unless multiple models contradict it). In MIMIC, STraTS’s binary predictions for CardiacInjury are extremely high prevalence (~55%), while all other voters are zero; majority vote still collapses to zero. This pattern is a *strong internal diagnostic* that CardiacInjury is not a valid treatment in your final dataset (either the rule-based definition never triggers due to missing labs/threshold mismatch, or there is a mapping/thresholding bug; and at least one model’s post-threshold behavior is inconsistent with the reference).

This is not a minor detail: it explains why your CATE pipeline cannot produce CardiacInjury effects and also warns that **some tag columns may have label definition problems** that should be audited before drawing substantive conclusions for that treatment.

Mortality prediction using latent variables

Your “latent validation” step trains mortality predictors using the latent tags as features. This is an important internal check: if latent tags are clinically meaningful, they should carry predictive information for mortality.

From your output logs:

- **PhysioNet** mortality prediction from latent tags:
 - Logistic regression test AUROC ≈ 0.735 ; test AUPRC ≈ 0.312
 - MLP test AUROC ≈ 0.747 ; test AUPRC ≈ 0.338
- **MIMIC** mortality prediction from latent tags:
 - Logistic regression test AUROC ≈ 0.762 ; test AUPRC ≈ 0.304
 - MLP test AUROC ≈ 0.765 ; test AUPRC ≈ 0.313

Interpretation:

- AUROC ~ 0.74 – 0.77 indicates **meaningful but incomplete** risk stratification. In ICU medicine, organ dysfunction scores (like SOFA) are predictive of mortality; Sepsis-3 explicitly ties a SOFA increase ≥ 2 to a nontrivial in-hospital mortality risk. ¹³ Thus, it is expected that a set of organ dysfunction tags has predictive value.
- The fact that performance is **not extremely high** is actually compatible with your causal goal: if mortality were almost deterministically implied by the tags, it would raise stronger concerns about label leakage or near-outcome encoding.
- However, moderate discriminative performance also implies **substantial residual heterogeneity** in mortality risk that is not captured by the tags—consistent with the possibility of **residual confounding** in downstream causal estimates.

This is a reasonable “incentive signal” from the data: your tags look like coarse organ dysfunction summaries that behave like coarse organ dysfunction summaries should.

CATE estimates and medical plausibility checks

What your CATE summaries show (risk-difference scale)

Your CATE module outputs **mean CATE** as an *absolute risk difference* in in-hospital mortality (e.g., $+0.125 \approx +12.5$ percentage points). Below I interpret the top results for each dataset, and I intentionally contrast:

- **Unadjusted association** (computed from your summary rates: $P(Y=1|T=1) - P(Y=1|T=0)$)
- **Matching estimate** (your matched-pair effect summary)
- **CausalForestDML mean CATE** (your main “causal ML” estimate)

This triangulation is essential because it directly addresses your request: “correlation can be high but not causal,” and it highlights where the model departs from raw association.

PhysioNet cohort (baseline mortality $\approx 14.24\%$)

Your largest positive mean CATEs are:

- **RenalFail**: mean CATE $\approx +12.5$ pp (unadjusted RD $\approx +16.0$ pp; matching $\approx +14.4$ pp)
- **HepFail**: mean CATE $\approx +10.1$ pp (unadjusted RD $\approx +11.8$ pp; matching $\approx +12.8$ pp)
- **Severity**: mean CATE $\approx +9.2$ pp (unadjusted RD $\approx +11.2$ pp; matching $\approx +10.8$ pp)
- **Metab**: mean CATE $\approx +8.7$ pp (unadjusted RD $\approx +12.7$ pp; matching $\approx +10.7$ pp)

Two tags are **negative** in the CausalForestDML summary:

- **Shock**: mean CATE ≈ -0.6 pp (but unadjusted RD $\approx +6.7$ pp, matching $\approx +3.8$ pp)
- **HemeFail**: mean CATE ≈ -1.8 pp (unadjusted RD $\approx +1.7$ pp, matching ≈ -0.6 pp)

These two negative estimates are the most important “be critical” signals in the PhysioNet run.

MIMIC cohort (baseline mortality $\approx 12.08\%$)

Largest mean CATEs:

- **HepaticDysfunction**: mean CATE $\approx +15.5$ pp (unadjusted RD $\approx +21.5$ pp; matching $\approx +19.8$ pp)
- **Severity**: mean CATE $\approx +10.0$ pp (unadjusted RD $\approx +11.8$ pp; matching $\approx +11.4$ pp)
- **Inflammation / RenalDysfunction / CoagDysfunction / MetabolicDerangement**: mean CATEs cluster around **+6.1 to +6.8 pp**

Shock is **positive but small**:

- **Shock**: mean CATE $\approx +2.2$ pp (unadjusted RD $\approx +11.1$ pp; matching $\approx +4.7$ pp)

So, across datasets, “renal” and “hepatic” are consistently strong, while “shock” collapses dramatically after adjustment (especially in MIMIC).

Medical literature plausibility: do the directions and magnitudes make sense?

Below are targeted “justifications” grounded in published clinical evidence. I focus on *direction* and *order-of-magnitude plausibility*, as you requested (not exact numerical matching).

Renal dysfunction → higher mortality is strongly plausible

Renal dysfunction in your tags closely matches SOFA renal dysfunction thresholds (creatinine ≥ 2 mg/dL and/or low urine output). ³ AKI/renal dysfunction is widely associated with increased mortality in critically ill cohorts.

Examples from the literature supporting the incentive:

- ICU studies evaluating AKI in critically ill patients (using KDIGO-style criteria) show AKI is associated with increased mortality risk and that severity/duration matters. ¹⁴
- In PhysioNet you estimate +12.5 pp (baseline 14.2%). A risk-difference of this magnitude is not obviously out of range for a cohort where mortality among renal-failure-tagged patients is materially higher than those without early renal dysfunction.

Thus, **RenalFail/RenalDysfunction effects are among the most defensible outputs in your CATE tables**, directionally and conceptually.

Hepatic dysfunction (bilirubin ≥ 2) → higher mortality is plausible and aligns surprisingly well

Your strongest MIMIC effect is HepaticDysfunction (+15.5 pp). This aligns with ICU literature showing bilirubin elevations (including thresholds around ≥ 2 mg/dL) are associated with higher mortality.

For example:

- A retrospective ICU study reported that bilirubin ≥ 2 mg/dL is associated with substantially higher hospital mortality and remains associated after multivariable adjustment; the crude mortality differences in that paper are on the order of tens of percentage points. ¹⁵
- A larger critical care observational literature notes hyperbilirubinemia is common and associated with worse outcomes; some analyses report roughly linear mortality increases over bilirubin ranges in the low-to-moderate region. ¹⁶

So, the *direction* and the *large magnitude relative to baseline mortality* for hepatic dysfunction is clinically believable as an “organ dysfunction state increases death risk” signal.

Coagulation/hematologic dysfunction: expected positive association, but your PhysioNet sign is a warning

SOFA's coagulation component uses platelet thresholds (e.g., platelet count $< 100 \times 10^3$ /mCL) as meaningful dysfunction levels. ³ Thrombocytopenia in ICU patients is widely associated with increased mortality in observational cohorts and reviews. ¹⁷

This makes your **MIMIC CoagDysfunction** positive mean CATE (+6.3 pp) directionally plausible.

However, **PhysioNet HemeFail** is negative (−1.8 pp), despite:

- unadjusted association being slightly positive (+1.7 pp), and
- broad critical care literature supporting thrombocytopenia as a marker of severity and worse outcomes. ¹⁷

This negative sign is **not impossible** (it could arise from conditioning/overadjustment or label definition quirks), but it *is* exactly the sort of result that should prevent you from making a confident medical statement like “hematologic failure reduces mortality.” The most likely interpretations are methodological (see below), not physiologic.

Metabolic derangement / acidosis → higher mortality is plausible

ICU metabolic acidosis (including thresholds like pH < 7.30) has been associated with higher ICU and hospital mortality in large observational datasets and reviews. ¹⁸ Lactate-related and other metabolic acidoses are repeatedly linked with worse outcomes. ¹⁹

This supports the consistent positive signals you see (PhysioNet +8.7 pp; MIMIC +6.1 pp).

Respiratory dysfunction: direction is right; magnitude is plausible as a coarse early indicator

Respiratory dysfunction indicators derived from $\text{PaO}_2/\text{FiO}_2$ are foundational in ARDS definitions and prognostication. ²⁰ The $\text{PaO}_2/\text{FiO}_2$ ratio associates with outcomes in sepsis/ARDS cohorts, though sometimes with only moderate standalone prognostic power. ²¹

Your RespFail positive effects (PhysioNet +4.4 pp; MIMIC +5.0 pp) are directionally consistent with this literature.

Neurologic dysfunction: direction is plausible

GCS is embedded in SOFA scoring and is known to be strongly associated with ICU mortality in large ICU cohorts and severity models. ²² Studies have specifically highlighted that GCS can dominate associations between SOFA and mortality in mixed ICU populations. ²³

Thus your positive NeuroFail/NeuroDysfunction effects (PhysioNet +4.5 pp; MIMIC +4.5 pp) are plausible.

Shock: your “high correlation but small CATE” pattern is plausible—but not for the reason “shock isn’t causal”

Your own numbers demonstrate precisely the phenomenon you described:

- **MIMIC:** unadjusted RD for Shock is about **+11.1 pp**, but CATE is only **+2.2 pp**.
- **PhysioNet:** unadjusted RD is about **+6.7 pp**, but CATE is slightly **negative (−0.6 pp)**, while matching remains positive (+3.8 pp).

Clinically, shock (especially septic shock as defined by Sepsis-3 physiology plus vasopressor requirement) is associated with very high mortality, often cited as >40% for the classic septic shock phenotype in Sepsis-3 derivation/validation discussions. ²⁴ Shock is also central in guideline discussions of perfusion thresholds (MAP ~65) and lactate-guided resuscitation. ²⁵

So, if shock were a clean, early, well-defined binary exposure, you would typically expect a positive effect.

Why can your *adjusted* CATE be small or even negative?

Two evidence-backed causal-inference mechanisms can plausibly explain it:

1. **Conditioning/adjustment can induce collider or selection bias** if your adjustment set includes variables that are common effects of shock and mortality (or their causes), or otherwise opens noncausal paths. Collider-stratification bias is a well-described phenomenon in epidemiology and DAG-based reasoning. ²⁶
In ICU settings, “severity” constructs and organ dysfunction variables can easily play complicated roles (confounder vs mediator vs collider) depending on temporal ordering and measurement definitions.
2. **“Shock” may violate the “single well-defined treatment version” idea** (multiple versions of the shock construct), leading to unstable or mixed counterfactual meaning. ²⁷

Importantly, this does **not** justify the claim “shock is not the cause of death.” Rather, it indicates: **your statistical estimand for “Shock” after your particular adjustment strategy appears to represent a**

narrow incremental contrast (shock versus no shock *among patients matched/adjusted on correlated severity and organ dysfunction features*), not the full physiologic harm of shock.

This is exactly the kind of nuance your output supports: **high association but weak/unstable incremental effect** is a credible outcome in observational settings even when the underlying syndrome is lethal.

Validation and credibility checks: permutation tests and sensitivity outputs

Permutation tests: strong sanity check, but not a proof of causality

Your permutation tests operate like placebo/falsification checks: if you break the treatment–outcome relationship by permuting labels, a sensible effect estimator should return effects near zero. This resonates with the broader “negative control/falsification” philosophy in observational causal inference: falsification tests can detect residual bias mechanisms, but a passing test does not prove identification.

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What your outputs show (representative examples):

- **PhysioNet RenalFail**: real mean CATE $\approx +0.125$; when permuting treatment labels, mean permuted CATE $\approx +0.001$ (SD ≈ 0.009); z-scores are large, indicating the real estimate is far from the permutation null.
- **MIMIC HepaticDysfunction**: real mean CATE $\approx +0.155$; treatment-permutation mean $\approx +0.001$ (SD ≈ 0.004).

Across most treatments, **permuted mean effects cluster near zero** and z-scores are large for the strongest effects. This is a meaningful credibility signal about your *pipeline mechanics*: it suggests the estimator isn’t merely producing large effects as an artifact of model misspecification unrelated to label alignment.

However, what permutation *cannot* address is the core observational problem: if an unmeasured confounder drives both the true treatment and outcome, permuting labels destroys that structure and will still yield near-zero effects—even though the real-world estimate may be confounded. So: **permutation tests support internal validity against “spurious estimator behavior,” but not identification against confounding.**

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Sensitivity analysis and robustness values: how to read your “RV” numbers

Your analysis outputs include robustness/sensitivity metrics (e.g., RV and sensitivity intervals). The concept of a “robustness value” is well-established in sensitivity analysis: it describes the minimum strength of unobserved confounding (in a defined sense) that would be needed to alter qualitative conclusions (e.g., move an estimate to include 0).

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In your outputs:

- **Shock** tends to show **RV ≈ 0** and sensitivity intervals that cross 0, which is consistent with a weak/unstable estimated effect where little confounding strength is needed to nullify the conclusion.

- **Renal and hepatic dysfunction** frequently show **non-zero RV** and sensitivity intervals that remain positive in the reported setting—consistent with being among the more robust signals in your run.

Two cautions are necessary:

1. Sensitivity analysis is always conditional on (i) a chosen sensitivity model/parameterization and (ii) the assumption that the adjustment set and modeling choices are otherwise valid. ²⁹
2. The robustness value is not a statement that unmeasured confounding is absent; it is a statement about **how strong** such confounding would need to be to overturn the inference, under the framework's assumptions. ²⁹

So RV is best used comparatively: **which treatments look fragile vs durable** relative to others within the same pipeline.

Critical appraisal: what your results support, and what they do not

What you *can* reasonably conclude from the combined evidence

Across both datasets, the following pattern looks **internally consistent** and **externally plausible**:

- **Early renal and hepatic dysfunction tags have large positive estimated effects on mortality risk** (CATE and matching estimates), and these align with organ dysfunction scoring logic (SOFA thresholds) and published ICU outcome associations for AKI and hyperbilirubinemia. ³⁰
- **Metabolic derangement and broad severity constructs show positive effects**, consistent with ICU evidence that acid-base derangements and higher organ dysfunction burden correlate with worse outcomes. ³¹
- **Permutation tests behave as you want them to behave**: when you destroy the treatment/outcome mapping, mean effects revert near zero, supporting that your estimation stack is not producing strong effects under random labels. ²⁸

These support your “incentive from the data” goal: there is a coherent signal that *organ dysfunction states* are connected to mortality in plausible directions, and the causal learner produces estimates consistent with that general medical knowledge.

The biggest red flags that should constrain any thesis-level causal claims

1. Shock sign instability (and negative Shock in PhysioNet CausalForestDML)

Given the clinical severity of shock, a negative average effect is difficult to accept as a physiologic causal claim. The more defensible interpretation is methodological: overadjustment, collider bias, multiple treatment versions, and/or mismatch between “Shock tag” and “septic shock as a causal exposure.” ³²

Your matching estimate for Shock stays positive in both datasets, and unadjusted association is strongly positive—so the negative sign is estimator/adjustment-sensitive. That means: **you should not present a clean “Shock is not causal” narrative**. The truthful narrative is: *shock is highly associated; incremental adjusted effect is small/unstable in this pipeline; interpretation depends critically on adjustment validity and treatment definition.*

2. CardiacInjury is not estimable in your final cohort

With a rule-based prevalence of 0%, this treatment is essentially undefined for causal estimation (no variation). Any narrative about CardiacInjury CATE should be explicitly marked “not

estimable / label construction failed,” and you should connect that to known troponin prognostic literature only as a motivation for fixing the label—not as validation of your current estimate. (For context: elevated troponin in critical illness is associated with increased mortality in systematic reviews, but your current pipeline cannot leverage that because the tag has no usable positives.) ³³

3. High treatment prevalences suggest coarse, overlapping syndromic tags

Many tags have prevalence 0.6–0.8. In such regimes, overlap between tags is likely high (e.g., Severity/Inflammation/Shock/RespFail co-occur), making adjustment sets and positivity/overlap more delicate. This can inflate estimator sensitivity to modeling choices and raise the plausibility of collider/mediator misclassification in DAG-based adjustment.

4. Passing permutation tests does not resolve unmeasured confounding

This is not a weakness of your work; it’s just a limitation of what permutation/placebo tests can do. They help detect gross pipeline artifacts, but they do not prove ignorability. ²⁸

A truth-anchored way to phrase your key examples in the thesis

These phrasing templates stay loyal to what your results and the literature jointly support:

• **Renal dysfunction example (PhysioNet)**

“In the PhysioNet cohort, the model estimates that early renal dysfunction (SOFA-consistent creatinine/urine output patterns) is associated with an increase in mortality of roughly 10–15 percentage points under the pipeline’s adjustment strategy. This direction is consistent with ICU evidence that AKI is associated with increased mortality and worse outcomes in critically ill patients.” ³⁴

• **Hepatic dysfunction example (MIMIC)**

“In the MIMIC cohort, hepatic dysfunction is among the largest estimated effects. This aligns with literature showing bilirubin elevations (including ≥ 2 mg/dL thresholds) are independently associated with higher hospital mortality in ICU populations.” ³⁵

• **Shock example (correlation vs incremental ‘causal’ estimate)**

“Shock has one of the strongest raw associations with mortality, yet the estimated incremental average effect after adjustment is small (and in one dataset slightly negative) in the causal forest outputs. This is compatible with known causal-inference phenomena: adjustment can change estimates substantially in observational data, and conditioning on colliders or ill-defined treatment versions can yield attenuated or unstable effects. Therefore, these results are better interpreted as an indicator of estimator/adjustment sensitivity than as evidence that shock is ‘not causal.’” ³²

These statements are defensible, evidence-linked, and do not oversell causality.

¹ ² ²⁷ <https://pmc.ncbi.nlm.nih.gov/articles/PMC4219328/>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC4219328/>

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